Ivan Chen

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EDUCATION

Rochester Institute of Technology 2024 – 2028
Bachelor of Science in Computer Engineering Technology | GPA: 3.68/4.0

Relevant Coursework: Control Systems, HDL, Microcontroller Systems, Digital Systems Design, Advanced Electronics

EXPERIENCE

Rivian and Volkswagen Group Technologies Irvine, CA
Software Engineering Intern May 2026 – August 2026

- Engineered a platform agnostic modeling framework and gray box system identification methodology for Software-in-the-Loop testing of EV inverters, achieving 96% model fidelity in normal and current limited regimes.
- Implemented and validated a real-time active speed limiting vehicle feature in C through SIL and in-vehicle testing.
- Built a RESTful API-driven vehicle data manipulation tool with a GUI for users and MCP server for LLM agents.
- Performed root cause analysis on bug tickets and assisted with vehicle tests and resolution solution tuning.

NYC Mayor's Office of Management and Budget New York, NY
Software Engineering Intern June 2025 – August 2025

- Automated end-to-end tax data pipelines with automated report generation/notification with MS Power Automate.
- Reduced daily data cleaning / processing time by 93% to 1.5 minutes, using Pandas and Python.
- Architected internal tax legislation research LLM + RAG system, with ChromaDB and LLaMA.

StuyPulse, FIRST Robotics New York, NY
Vice President of Software Engineering October 2020 – June 2024

- Developed a robot localization system (1" accuracy) using AprilTags + OpenCV on headless Linux coprocessors.
- Engineered PID+FF control systems for mechanisms on 8 highly performing robots using Java, including a double-jointed arm, telescoping mechanism, a color-sorted conveyor system, and autonomous driving systems.
- Led 50+ and taught over 200+ members in software and robotics topics and managing interdepartmental relations.
- Perform root cause analysis on software/hardware issues for 48+ teams; presented workshops to 400+ attendees.

PROJECTS

Autonomous Race Car Development with Simulation **RIT Formula Racing Team**

- Develop a high-fidelity multi-threaded vehicle dynamics simulation in C using a bicycle model and torque vectoring to analyze and test a full perception, planning, and control pipeline.
- Engineer a 3D perception stack to process simulated LiDAR frames, sent via UDP, into point clouds with detection and classification algorithms like RANSAC ground plane removal and A-BIRCH for cone clustering.
- Develop a multi-layer perceptron for cone classification using reflectivity and point cloud data.
- Implement a Stanley vehicle controller for trajectory tracking, migrating from previous pure pursuit approach.

Autonomous Glider System **RIT Aero Design Team**

- Developed C drivers for interfacing with inertial sensors, GPS, and servo motors on control surfaces.
- Implemented a high-level finite state machine to move between control states such as turning, landing, etc.
- Applied Unscented Kalman Filter (UKF) with sensor data and PID control for autonomous glider navigation.

SKILLS

- **Technical Skills**: C/C++, Python, VHDL, MSP432, Robotics, Computer Vision, Firmware, Embedded Systems, Motion Planning, Sensors (LiDAR, Cameras, IMUs, Encoders, GPS), Classical Controls, React.js, SQL, Statistics
- **Tools**: CANape, Quartus Prime, ModelSim, Multisim, Oscilloscopes, OpenCV, Fusion360, Git, Docker, MS Excel